

Reproducibility and label leakage in machine learning on Q-CHAT-10 toddler screening data: a synthetic benchmark of a 1D CNN and classical classifiers

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Abstract

Background: Machine-learning (ML) studies on the public Q-CHAT-10 toddler autism screening dataset routinely report near-perfect accuracy. In that dataset, however, the class label is not a clinical diagnosis. It is generated by the questionnaire's own rule (item sum > 3), so the reported performance measures how well a model reproduces a threshold-defined screening label. **Methods:** We ran a methodological study on a fully specified synthetic benchmark (1054 records), constructed to reproduce the public dataset's schema, class balance and deterministic label rule. The published scoring rule applied directly, with no learning, served as the reference. A one-dimensional convolutional neural network (1D CNN, 76,161 parameters) and seven classical classifiers were compared with it under a leakage-controlled pipeline, which drops the aggregate score and fits all encoders on training data only. Evaluation used 20 repeated stratified 80/20 holdout splits. On a held-out split we computed Wilson confidence intervals, exact McNemar tests and positive predictive values (PPV) across illustrative prevalences. We also ran a feature-subset ablation, permutation importance, 10 CNN re-initialisations and a CNN feature-order sensitivity analysis. **Results:** The direct rule and logistic regression both reproduced the label without error on every split (accuracy 1.000 and 1.000 ± 0.000). The 1D CNN reached 0.997 ± 0.004 (AUC 1.000) and did not improve on it. Tree-based and probabilistic models were less accurate. Reordering the CNN's input columns changed its mean accuracy by less than one percentage point. With demographic and history attributes alone, all models were near chance for this label (AUC ≤ 0.531). Even for the error-free models, the lower 95% bound of specificity (0.945 on 66 negatives) implies a PPV as low as 0.354 at an illustrative 3% prevalence. **Conclusions:** On this benchmark, near-perfect ML performance shows that the scoring rule is learnable. It does not show diagnostic or screening validity, and neither a linear model nor a 1D CNN improves on applying the rule directly. We recommend that Q-CHAT-10 ML studies report the label's provenance, remove the score, compare against direct scoring, and validate against clinician-confirmed diagnoses. All numerical results here are from synthetic data.

Keywords: Autism spectrum disorder, Q-CHAT-10, Toddler screening, Label leakage, Reproducibility, Convolutional neural network, Synthetic benchmark

1. Introduction

Autism spectrum disorder (ASD) is a lifelong neurodevelopmental condition characterised by persistent differences in social communication and interaction, together with restricted and repetitive patterns of behaviour [1]. The most recent US surveillance estimated that 1 in 31 children aged eight years were identified with ASD in 2022 [2]. Early intensive behavioural intervention in toddlerhood improves cognitive, language and adaptive outcomes [3]. This makes accurate and scalable screening in the first three years important. Short parent-report instruments such as the M-CHAT-R/F [4] and the Q-CHAT-10 [5] are used as first-line screens.

Machine learning (ML) has been applied to autism instruments to shorten them, to fuse information across them, and to automate decisions [6, 7, 8, 9]. A widely used public resource is

the toddler screening dataset collected with the ASDTests mobile application [10, 11]. On this and related datasets, many studies report accuracies close to 100% [14].

Such results need careful interpretation. In the toddler dataset, the class label "ASD traits" is not an independent clinical diagnosis. It is assigned when the Q-CHAT-10 item sum exceeds 3, which is the instrument's own referral rule [5]. A model given the ten items can therefore reproduce the label exactly. High accuracy on this benchmark measures agreement with the scoring rule, not the ability to identify autism. The dataset also contains the aggregate score column. Using it as a feature, or fitting encoders on the full dataset before splitting, leaks label information [15]. Throughout this paper we therefore use *threshold-defined screening label* for the prediction target and reserve "diagnosis" for clinician-confirmed status.

This paper is a methodological and reproducibility study. Because the original data file could not be obtained in our computing environment, all experiments were run on a synthetic benchmark constructed to reproduce the dataset's schema, class balance and deterministic label rule. It is not a validation study

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on the public dataset. The generator is specified in full in Section 3.2. The contributions are:

- a leakage-controlled, fully scripted pipeline in which the aggregate score is removed and all transformations are fitted on training data only;
- a comparison of the published scoring rule applied directly, a compact 1D CNN and seven classical classifiers over 20 repeated stratified holdout splits, with confidence intervals, paired significance tests and predictive values across illustrative prevalences;
- a feature-subset ablation, permutation importance with uncertainty, CNN re-initialisation and CNN feature-order experiments, which show which inputs the models use to reproduce this label and how stable the results are;
- reporting recommendations for ML studies that use threshold-defined screening labels.

Section 2 reviews related work. Section 3 describes the data, the synthetic generator, the models and the protocol. Section 4 reports the results. Section 5 discusses them, and Section 6 concludes.

2. Related work

ML for shortening and automating autism instruments. Wall et al. [6] used alternating decision trees to reduce ADOS Module 1 to a few items with little loss of accuracy. Kosmicki et al. [7] used feature selection to find minimal behaviour sets for autism detection. Duda et al. [9] applied ML to distinguish autism from ADHD using the Social Responsiveness Scale. Bone et al. [8] examined ML-derived instruments critically. They showed how sampling choices and label definitions can make results look better than they are, and they proposed fusing information across instruments.

Q-CHAT-10 and the ASDTests datasets. Allison et al. [5] derived the Q-CHAT-10 from the full Q-CHAT. In their sample of 126 toddler cases and 754 controls, a cut-point of 3 gave a sensitivity of 0.91, a specificity of 0.89 and a positive predictive value (PPV) of 0.58. The published instrument advises considering referral when a child scores more than 3. Thabtah [10] discussed ML adaptation of ASD screening in relation to DSM-5, and later described the ASDTests application and its behavioural datasets [11]. Thabtah and Peebles [12] proposed a rule-induction model for autism detection. Both *Health Informatics Journal* articles [11, 12] are the subject of a 2021 expression of concern about their peer review [13]. We cite them only as the source and description of the dataset. Akter et al. [14] compared several classifiers and feature transformations on the ASDTests toddler, child, adolescent and adult datasets and reported very high accuracies.

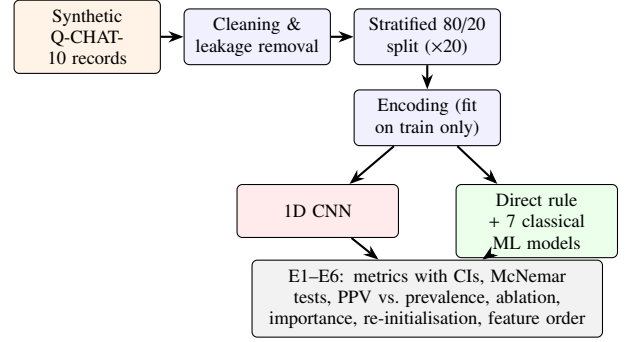


Figure 1: Overview of the experimental pipeline.

Scope of the evidence. We separate four things. The first is *dataset provenance*: the public toddler records, collected with the ASDTests application and described in [11]. The second is *claims made in the original articles*, which we do not re-examine and which are under an expression of concern [13]. The third is *independent validation of the dataset*: we know of none that confirms its labels against clinical diagnosis. The fourth is *the present experiment*, which uses only a synthetic benchmark. Our conclusions rest on the label’s published definition and on the synthetic experiment, not on the other three.

Deep learning for ASD. Heinsfeld et al. [16] applied deep learning to resting-state fMRI connectivity from the ABIDE consortium. One-dimensional CNNs are well established for sequential and signal-like inputs [17]. On generic tabular data, however, deep models do not reliably outperform tree ensembles [18]. This motivates evaluating a CNN against strong classical baselines under an identical protocol.

3. Materials and methods

Fig. 1 summarises the workflow.

3.1. Reference dataset and label definition

The reference dataset is the public *Autism Screening Data for Toddlers* [11]. Its records were collected with the ASDTests mobile application from parents, carers and health professionals of toddlers aged 12–36 months, and they contain no direct identifiers. Each record holds the ten Q-CHAT-10 items (A1–A10), age in months, sex, ethnicity, history of neonatal jaundice, a family member with ASD, the respondent type, the Q-CHAT-10 score and the class label. The items are coded so that 1 denotes an answer consistent with an ASD trait. The class label is “ASD traits” when $\sum_{j=1}^{10} A_j > 3$ and “No ASD traits” otherwise. This is the instrument’s referral rule, not a clinical diagnosis. The descriptor articles [11] do not report clinical confirmation of the labels.

3.2. Synthetic benchmark

The original CSV could not be downloaded in the computing environment used for this study. We therefore constructed a synthetic benchmark with the same size, schema,

Table 1: Generative parameters of the synthetic benchmark ($n = 1054$, seed 42). z : latent status. “Indep.”: independent of z .

Variable	If $z = 1$	If $z = 0$
Latent status z	Bernoulli(0.69)	
Item prob. p_{ij} (per item)	$U(0.45, 0.85)$	$U(0.03, 0.30)$
Items A_{ij}	Bernoulli(p_{ij}), independent given p_{ij}	
Sex (male)	0.74	0.60
Neonatal jaundice	0.30	0.22
Family member with ASD	0.18	0.12
Age (months)	Indep.; uniform integer 12–36	
Ethnicity (11 categories)	Indep.; uniform	
Respondent (5 categories)	Indep.; (0.93, 0.03, 0.01, 0.02, 0.01)	
Label	“ASD traits” iff $\sum_j A_{ij} > 3$	

approximate class balance and deterministic label rule, using `asd.data.make_synthetic` with seed 42. It is a benchmark built to reproduce these structural properties, not a statistical surrogate of the public data’s joint distribution. **All numerical results in this paper were obtained on this benchmark, not on the public dataset.** The benchmark was generated once, and all splits, seeds and experiments use the same 1054 records. Table 1 lists every generative parameter. Each toddler i receives a latent status $z_i \sim \text{Bernoulli}(0.69)$. Each item has its own response probability, $p_{ij} \sim U(0.45, 0.85)$ if $z_i = 1$ and $p_{ij} \sim U(0.03, 0.30)$ otherwise, and $A_{ij} \sim \text{Bernoulli}(p_{ij})$. Items are therefore independent given z_i and p_{ij} , and they are correlated with each other only through z_i . Sex, jaundice and family history depend on z_i with the small effect sizes in Table 1. Age, ethnicity and respondent type are independent of everything else. The label is computed from the items with the published rule, *not* from z_i . The label therefore differs from the latent status for some toddlers, and demographic attributes relate to the label only indirectly through z_i .

The benchmark matches the structural property that drives our conclusions: the label is a deterministic function of the items. It does not reproduce the true joint distribution of the public data, including real inter-item correlations, real demographic associations and data-entry artefacts. Absolute numbers, especially those for demographic features, reflect the generator’s parameter choices. They must be re-estimated on the public data, which the released script does with a single command.

The benchmark contains 1054 toddlers: 722 (68.5%) with the ASD-traits label and 332 without. 72.6% are male, mean age is 23.9 ± 7.1 months, 28.2% have a history of jaundice and 15.8% report a family member with ASD. Fig. 2 shows the proportion of trait answers per item in each class. Fig. 3 shows the score distribution, where the classes separate exactly at the threshold.

3.3. Preprocessing and leakage control

The case identifier is removed. The Q-CHAT-10 score is also removed, because it is the sum of A1–A10 and defines the label directly. With the score included, a single threshold on that one column classifies every record correctly. Categorical strings are normalised (trimmed and lower-cased). Sex, jaundice and family history are coded as binary variables. Ethnicity and respondent type are one-hot encoded, and categories unseen in training

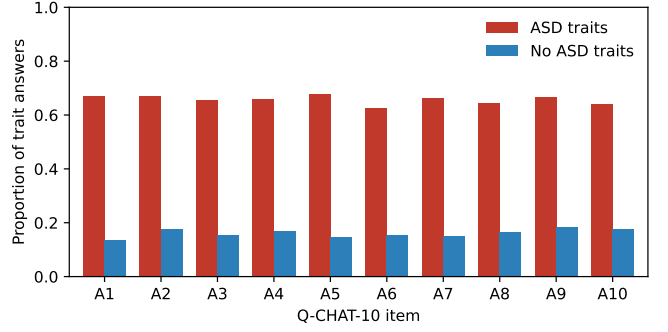


Figure 2: Proportion of trait-indicating answers to each Q-CHAT-10 item by label class (synthetic benchmark).

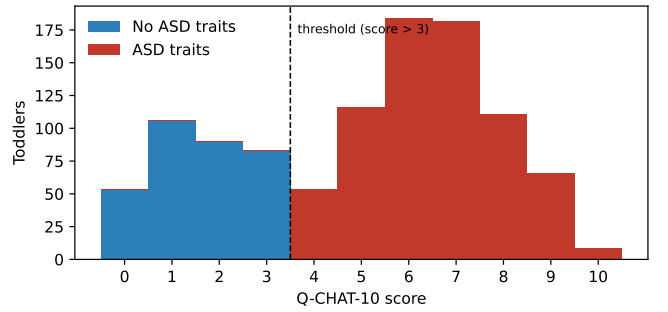


Figure 3: Distribution of the Q-CHAT-10 score (synthetic benchmark). The dashed line marks the rule (score > 3) that defines the label, and the two classes do not overlap across it.

are mapped to all zeros. Age is min–max scaled to $[0, 1]$, and the binary items pass through unchanged. The encoded vector has $d = 29$ features. Each split is stratified by class into 80% training and 20% test data. The encoder and scaler are fitted on the training portion only.

3.4. Direct scoring rule and classical machine-learning models

Direct Q-CHAT-10 rule. The reference baseline applies the published rule to the ten items, $\hat{y} = \mathbf{1}[\sum_j A_j > 3]$, with nothing fitted. For ROC analysis its score is the item sum. On a label defined by this rule it is correct by construction, so it marks the ceiling that any learned model can at best match.

Classical classifiers. Seven scikit-learn [19] classifiers were compared:

- logistic regression (LR), ℓ_2 -regularised with $C=1$;
- an RBF-kernel support vector machine (SVM, $C=1$) [20], with Platt scaling [21] fitted by internal cross-validation;
- k -nearest neighbours (KNN, $k=5$);
- a decision tree (DT) with a maximum depth of 6;
- a random forest (RF) [22] with 300 trees;
- gradient boosting (GB) [23] with 100 trees of depth 3;

Table 2: Layer-by-layer structure of the 1D CNN, taken from the executable model (`model.summary()`).

Layer (configuration)	Output shape	Parameters
Input	29×1	0
Conv1D (32, $k=3$, ReLU)	29×32	128
Conv1D (64, $k=3$, ReLU)	29×64	6,208
MaxPool1D (size 2)	14×64	0
Dropout (0.25)	14×64	0
Conv1D (64, $k=3$, ReLU)	14×64	12,352
Flatten	896	0
Dense (64, relu)	64	57,408
Dropout (0.3)	64	0
Dense (1, sigmoid)	1	65
Total (all trainable)		76,161

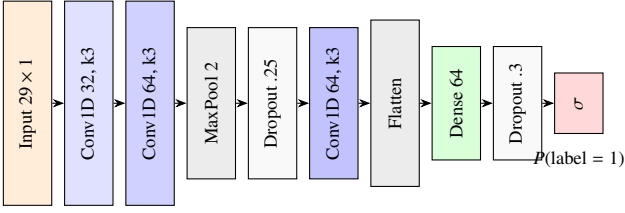


Figure 4: Architecture of the 1D CNN. ReLU follows every convolution and the hidden dense layer.

- Gaussian naive Bayes (NB).

Hyper-parameters were fixed a priori and not tuned. Model-internal randomness uses the split’s seed.

3.5. One-dimensional CNN

The encoded record $\mathbf{x} \in \mathbb{R}^d$ is treated as a one-channel sequence of length d . Because A1–A10 occupy adjacent positions, the convolution kernels can combine neighbouring items, and weight sharing keeps the parameter count small [17]. The remaining encoded variables (history flags, scaled age and one-hot categories) have no natural order, so their neighbourhoods are arbitrary. Experiment E6 therefore tests whether the column order matters. The architecture is shown in Fig. 4 and Table 2; the table is generated from the executable model. Flattening rather than global pooling keeps each feature’s position, which is informative because each position is a specific question. Max-pooling uses floor rounding, so the sequence length goes from 29 to 14, and the model has 76,161 trainable parameters.

The CNN is trained with Adam [25] (learning rate 10^{-3} , batch size 32) to minimise binary cross-entropy,

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N [y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)]. \quad (1)$$

A class-stratified 15% of each training set is held out for validation. The validation subset is drawn with the run’s seed, so it differs between runs. Dropout [24] is applied as in Table 2. The learning rate is halved after 5 epochs without improvement in validation loss. Training stops after 20 such epochs (maximum 150), and the weights with the lowest validation loss are

restored. Weights use the Keras defaults (Glorot-uniform kernels, zero biases). Python, NumPy and TensorFlow seeds are set per run, and TensorFlow’s deterministic-operation mode is enabled [26]. The CNN was trained 20 times in E2 (once per split), 3×20 times in E3, 10 further times in E5 and $(3 + 1) \times 5$ times in E6.

3.6. Experimental protocol

Six experiments were run. All seeds and commands are listed in Appendix A.

- **E1 (held-out split):** one fixed split (seed 42; 211 test records, 145 positive and 66 negative). It provides the confusion matrices, ROC curves, screening metrics with Wilson 95% confidence intervals [27], exact McNemar tests [28] comparing each model with the direct rule on the same test records, and PPV across illustrative prevalences.
- **E2 (repeated holdout):** the direct rule and all eight models on 20 repeated stratified 80/20 holdout splits of the same benchmark (seeds 0–19). These are repeated random partitions, not independent datasets. We report the mean, standard deviation and the 2.5th–97.5th percentile range across splits. Together the test sets cover 1045 of the 1054 records. The splits overlap, so they are not independent samples, and the SD describes split-to-split variability rather than a standard error.
- **E3 (ablation):** LR, RF and the CNN trained on (i) the ten items only, (ii) demographic and history attributes only, and (iii) all features, using the same 20 splits.
- **E4 (importance):** permutation importance on the test sets of the first 5 E2 splits, with 30 permutations per variable and split. Importance is the mean decrease in ROC-AUC, and one-hot columns of a variable are permuted together. We also report the LR coefficients from E1.
- **E5 (CNN stability):** the CNN retrained 10 times on the E1 split with seeds 1000–1009, which set both initialisation and validation subset. The E1 run (seed 42) is *not* one of these ten.
- **E6 (feature order):** the CNN retrained on the first 5 E2 splits with the encoded columns (i) in the original order (items first), (ii) reversed (items last), and (iii) in 3 random permutations. The permutations use fixed seeds and are shared across splits.

Metrics. With TP, TN, FP and FN the counts of true and false positives and negatives (positive = ASD-traits label), we report

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}, \quad \text{Sensitivity} = \frac{TP}{TP+FN}, \quad (2)$$

$$\text{Specificity} = \frac{TN}{TN+FP}, \quad \text{PPV} = \frac{TP}{TP+FP}, \quad \text{NPV} = \frac{TN}{TN+FN}, \quad (3)$$

$$\text{LR}^+ = \frac{\text{Sens}}{1-\text{Spec}}, \quad \text{LR}^- = \frac{1-\text{Sens}}{\text{Spec}}, \quad (4)$$

together with the F1-score and ROC-AUC, all at a 0.5 threshold. PPV and NPV depend on prevalence, which is 68.5% in

the data. We therefore also compute PPV at illustrative prevalences $\pi \in \{1, 3, 5, 10\}\%$ as $\text{Sens } \pi / [\text{Sens } \pi + (1 - \text{Spec})(1 - \pi)]$, from both the point estimates and the lower confidence bounds. These values are illustrative. The 3% figure is close to the prevalence of identified ASD among eight-year-olds [2], and the prevalence among toddlers presenting for screening depends on the setting and is not known here. We did not use DeLong’s test [29] because several AUCs equal 1.000, where the test is uninformative. Paired comparisons use McNemar’s test on the E1 predictions.

Computational environment. Experiments ran on CPU only (Intel Xeon Processor @ 2.10GHz, 4 cores; Linux 6.18.44) with Python 3.11.15, TensorFlow 2.21.0, scikit-learn 1.9.1, NumPy 2.4.6 and pandas 3.0.6. Exact package versions are pinned in `requirements-lock.txt`. The command `python experiments.py --synthetic` regenerates every table, figure and number in this paper. It also writes `results_summary.json`, a machine-readable file of all results.

4. Results

All results in this section are on the synthetic benchmark.

4.1. Comparison over repeated splits (E2)

Table 3 summarises E2. The direct rule is correct by construction (accuracy 1.000, AUC 1.000). LR matched it on every split (accuracy 1.000 ± 0.000). The 1D CNN reached 0.997 ± 0.004 (sensitivity 0.997, specificity 0.997, AUC 1.000), close to but not above LR. The RBF SVM followed. The tree ensembles kept sensitivity high but lost specificity, because axis-aligned splits approximate a threshold on a sum of ten binary variables only coarsely. Naive Bayes, whose independence assumption does not hold for the items given the label, also lost specificity. The single decision tree was weakest (accuracy 0.895).

4.2. Held-out screening metrics and paired tests (E1)

Table 4 and Figs. 5–6 report E1. Against the direct rule, the McNemar p -value is 1.000 for LR and 0.500 for the CNN. The confidence intervals are wide despite point estimates of 1.000, because the test set contains only 66 negatives. For the rule and LR, the lower Wilson bounds are 0.974 for sensitivity and 0.945 for specificity. Table 5 shows what these bounds imply across illustrative prevalences. At the lower bounds, PPV ranges from 0.152 at 1% prevalence to 0.663 at 10% (0.354 at 3%). Even zero observed false positives on 66 negatives therefore cannot establish a high PPV at low prevalence. For comparison, the instrument’s published accuracy [5] implies PPVs of 0.077, 0.204 and 0.479 at 1%, 3% and 10%. These PPVs are much lower than the near-perfect benchmark figures suggest.

4.3. CNN stability across initialisations (E5)

Across 10 re-initialisations on the E1 split, CNN test accuracy was 1.000 ± 0.000 (minimum 1.000). These ten runs (seeds 1000–1009) do not include the E1 run with seed 42, which made 2 errors on the same test set (Table 4). Whether the CNN misclassifies a few borderline records therefore depends on its initialisation and validation subset. Early stopping ended training between epochs 49 and 119. Fig. 7 shows the mean $\pm$ SD learning curves. Validation loss fell sharply within the first few epochs and then plateaued, and training and validation curves stayed close.

4.4. Sensitivity to input-column order (E6)

Table 6 compares CNN performance across column orders on the first 5 splits. Accuracy was 0.997 with the original order and 0.997 with the order reversed. Across the 3 random permutations it was 0.990 ± 0.006 (minimum 0.981, AUC 1.000). Scattering the items across the sequence lowered mean accuracy slightly, by less than one percentage point, and left AUC essentially unchanged. The CNN’s performance therefore depends only marginally on the items being adjacent. This is consistent with the flatten-and-dense head doing most of the work, and it gives little support to a claim that convolution over item neighbourhoods drives the result.

4.5. Feature-subset ablation (E3)

Table 7 shows that, for this threshold-defined label, the items carry the predictive signal. With the items alone, LR reached 1.000 and the CNN 1.000, matching the full-feature models. With only age, sex, ethnicity, jaundice, family history and respondent type, all three models were near chance (AUC: LR 0.531, RF 0.515, CNN 0.530). In this setting the CNN’s accuracy (0.686) is close to the proportion of the majority class, because it predicted the ASD-traits label for nearly every test record. Its F1-score is therefore inflated by class imbalance, which is why accuracy and F1 must be read together with AUC and specificity. Under the generator, demographic attributes depend on the latent status only weakly and on the label only indirectly (Table 1). This result therefore says nothing about the clinical value of these attributes.

4.6. Feature importance (E4)

Fig. 8a shows permutation importance with error bars (SD over 5 splits $\times$ 30 permutations). For all three models, every item has a positive importance. For the CNN it ranges from 0.004 to 0.012 (largest for A1). The largest absolute importance of any demographic or history variable, for any model, is <0.001 . The LR coefficients (Fig. 8b) tell the same story. The item coefficients all lie between 1.93 and 2.81, close to the equal weights of the scoring rule, while no demographic coefficient exceeds 0.46 in magnitude. These rankings describe how the models reproduce an additive rule on synthetic data. They should not be read as a clinical ordering of the items.

Table 3: Test-set performance over 20 repeated stratified 80/20 holdout splits of the synthetic benchmark: mean $\pm$ SD, and the 2.5th–97.5th percentile range of accuracy across splits. The first row is the published scoring rule applied directly, with no learning. The remaining rows are sorted by F1-score.

Model	Accuracy	Acc. range	Sensitivity	Specificity	F1	AUC
Q-CHAT-10 rule ($\sum A_j > 3$)	1.000 $\pm$ 0.000	1.000–1.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000
Logistic Regression	1.000 $\pm$ 0.000	1.000–1.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000
1D CNN	0.997 $\pm$ 0.004	0.991–1.000	0.997 $\pm$ 0.005	0.997 $\pm$ 0.006	0.998 $\pm$ 0.003	1.000 $\pm$ 0.000
SVM (RBF)	0.994 $\pm$ 0.006	0.981–1.000	0.995 $\pm$ 0.007	0.992 $\pm$ 0.010	0.996 $\pm$ 0.005	1.000 $\pm$ 0.000
Gradient Boosting	0.983 $\pm$ 0.011	0.967–1.000	0.990 $\pm$ 0.009	0.967 $\pm$ 0.028	0.988 $\pm$ 0.008	0.999 $\pm$ 0.001
Random Forest	0.980 $\pm$ 0.009	0.960–0.991	0.998 $\pm$ 0.004	0.941 $\pm$ 0.028	0.986 $\pm$ 0.007	0.999 $\pm$ 0.001
KNN	0.966 $\pm$ 0.016	0.936–0.991	0.961 $\pm$ 0.019	0.976 $\pm$ 0.024	0.975 $\pm$ 0.012	0.996 $\pm$ 0.003
Naive Bayes	0.951 $\pm$ 0.013	0.926–0.969	0.983 $\pm$ 0.012	0.882 $\pm$ 0.033	0.965 $\pm$ 0.009	0.981 $\pm$ 0.016
Decision Tree	0.895 $\pm$ 0.021	0.857–0.934	0.914 $\pm$ 0.020	0.853 $\pm$ 0.047	0.923 $\pm$ 0.016	0.922 $\pm$ 0.024

Table 4: Screening metrics on the E1 test split ($n = 211$: 145 positive, 66 negative) with Wilson 95% confidence intervals. PPV and NPV are at the dataset’s prevalence; PPV_{3%} re-weights the same sensitivity and specificity to a 3% illustrative population prevalence (point estimate / value at the lower CI bounds). p : exact McNemar test against the direct Q-CHAT-10 rule.

Model	Sensitivity (95% CI)	Specificity (95% CI)	PPV	NPV	LR+	LR–	PPV _{3%}	p
Q-CHAT-10 rule	1.000 (0.974–1.000)	1.000 (0.945–1.000)	1.000	1.000	∞	0.000	1.000 / 0.354	–
Logistic Regression	1.000 (0.974–1.000)	1.000 (0.945–1.000)	1.000	1.000	∞	0.000	1.000 / 0.354	1.000
SVM (RBF)	1.000 (0.974–1.000)	1.000 (0.945–1.000)	1.000	1.000	∞	0.000	1.000 / 0.354	1.000
1D CNN	0.986 (0.951–0.996)	1.000 (0.945–1.000)	1.000	0.971	∞	0.014	1.000 / 0.348	0.500
Gradient Boosting	1.000 (0.974–1.000)	0.924 (0.835–0.967)	0.967	1.000	13.2	0.000	0.290 / 0.154	0.062
Random Forest	1.000 (0.974–1.000)	0.894 (0.797–0.948)	0.954	1.000	9.4	0.000	0.226 / 0.129	0.016
KNN	0.979 (0.941–0.993)	0.924 (0.835–0.967)	0.966	0.953	12.9	0.022	0.286 / 0.150	0.008
Naive Bayes	0.986 (0.951–0.996)	0.773 (0.658–0.857)	0.905	0.962	4.3	0.018	0.118 / 0.079	<0.001
Decision Tree	0.924 (0.869–0.957)	0.742 (0.626–0.833)	0.887	0.817	3.6	0.102	0.100 / 0.067	<0.001

Table 5: Positive predictive value across illustrative prevalences. For the E1 estimates each cell gives the point estimate / the value at the lower Wilson bounds of sensitivity and specificity. “Sensitivity 0.91 and specificity 0.89 at the published cut-point [5], for reference.

Prevalence	Rule / LR (E1)	1D CNN (E1)	Published ^a
1%	1.000 / 0.152	1.000 / 0.149	0.077
3%	1.000 / 0.354	1.000 / 0.348	0.204
5%	1.000 / 0.482	1.000 / 0.476	0.303
10%	1.000 / 0.663	1.000 / 0.658	0.479

Table 6: CNN sensitivity to input-column order (E6): mean $\pm$ SD and minimum over the first 5 E2 splits. Random orders use fixed permutation seeds shared across splits.

Column order	Accuracy	Min. accuracy	AUC
Original (items first)	0.997 $\pm$ 0.004	0.991	1.000 $\pm$ 0.000
Reversed (items last)	0.997 $\pm$ 0.003	0.995	1.000 $\pm$ 0.000
Random permutation 1	0.989 $\pm$ 0.003	0.986	0.999 $\pm$ 0.000
Random permutation 2	0.988 $\pm$ 0.004	0.981	1.000 $\pm$ 0.001
Random permutation 3	0.993 $\pm$ 0.009	0.981	1.000 $\pm$ 0.000

5. Discussion

What near-perfect accuracy means here. The reference label is a deterministic function of the inputs, $y = 1[\sum_j A_j > 3]$. On binary item inputs, a linear decision boundary can reproduce this threshold classification exactly. This lets LR match the direct rule on every split, and the CNN approximates the same boundary closely. On this label, a learned model has no practical advantage over adding up the ten answers, which is exactly what the rule baseline does. Reported accuracy gains on the

Table 7: Feature-subset ablation (mean $\pm$ SD over 20 splits).

Model	Accuracy	F1	AUC
<i>Q-CHAT-10 items only</i>			
Logistic Regression	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000
Random Forest	0.981 $\pm$ 0.010	0.986 $\pm$ 0.007	0.999 $\pm$ 0.001
1D CNN	1.000 $\pm$ 0.001	1.000 $\pm$ 0.001	1.000 $\pm$ 0.000
<i>Demographics only</i>			
Logistic Regression	0.674 $\pm$ 0.013	0.804 $\pm$ 0.010	0.531 $\pm$ 0.040
Random Forest	0.597 $\pm$ 0.030	0.717 $\pm$ 0.025	0.515 $\pm$ 0.043
1D CNN	0.686 $\pm$ 0.004	0.814 $\pm$ 0.002	0.530 $\pm$ 0.042
<i>All features</i>			
Logistic Regression	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000
Random Forest	0.980 $\pm$ 0.009	0.986 $\pm$ 0.007	0.999 $\pm$ 0.001
1D CNN	0.997 $\pm$ 0.004	0.998 $\pm$ 0.003	1.000 $\pm$ 0.000

benchmark are gains in reproducing a published scoring rule. They are not gains in detecting autism. This conclusion follows from how the label is defined, not from the synthetic data. It therefore applies to the public dataset, and to any other dataset whose label is computed from the model’s inputs.

What this benchmark cannot establish. Because the target is constructed from the inputs, good performance here says nothing about generalisation to other populations, robustness to distribution shift, discrimination of clinically diagnosed ASD, calibration or real-world screening utility. This holds for every model in this study, including the CNN.

Role of the CNN. The CNN’s shared kernels over adjacent items and its dense head can approximate a weighted item sum.

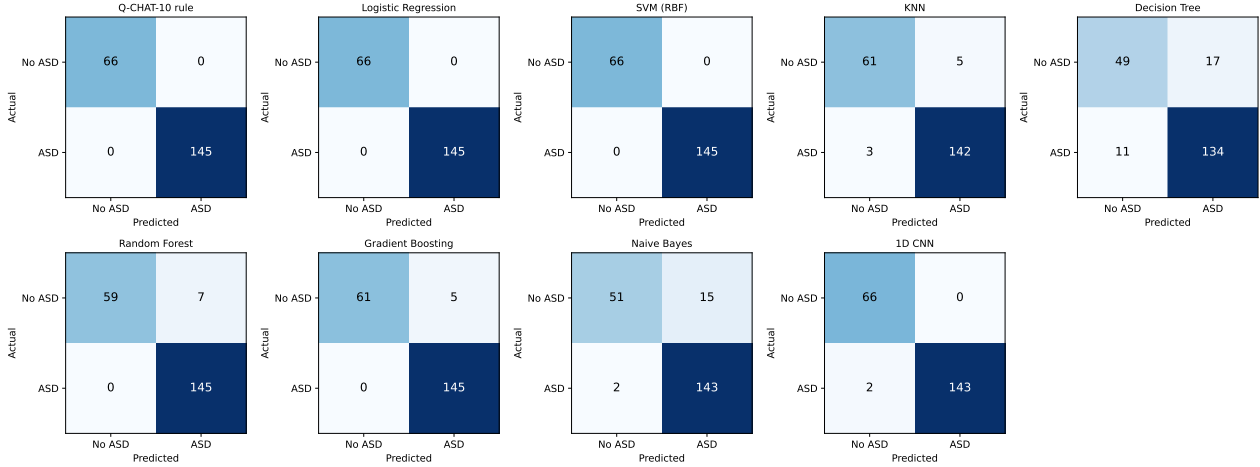


Figure 5: Confusion matrices on the E1 test split (seed 42; 211 records).

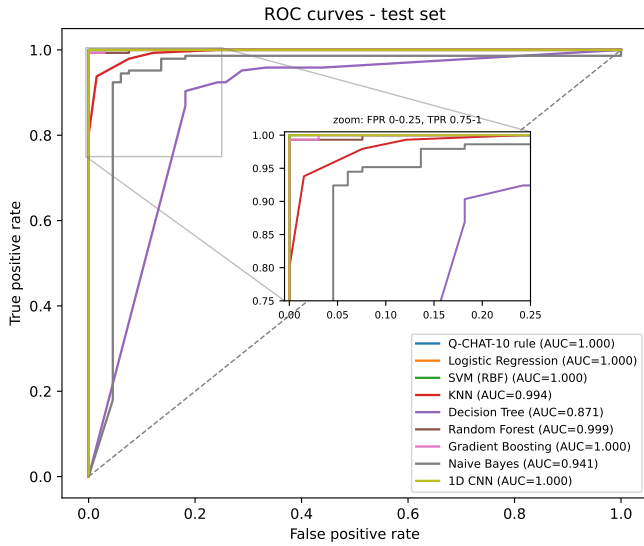


Figure 6: ROC curves on the E1 test split. The inset magnifies the region $FPR \leq 0.25$, $TPR \geq 0.75$, where the curves differ.

Dropout and early stopping keep it stable across splits and initialisations. However, LR matches its performance with about 30 parameters instead of 76,161, trains in milliseconds, and exposes coefficients that can be checked directly against the scoring rule. The direct rule needs no parameters at all. E6 also shows that the CNN’s result depends only marginally on the local item structure its convolutions were meant to exploit. Evaluating CNNs would be more informative with richer longitudinal or multimodal inputs, such as item response times, free-text answers or video, and an independently established clinical endpoint.

Leakage and reporting. With the score column retained, one threshold on that column classifies every record correctly. We recommend that ML studies on questionnaire-derived labels (i) state how the label was generated, (ii) list the input columns

and confirm that label-defining aggregates are excluded, (iii) state where preprocessing was fitted, (iv) report direct scoring of the instrument as a baseline, as in Table 3, and (v) report confidence intervals and prevalence-adjusted predictive values rather than accuracy alone.

Towards clinically meaningful evaluation. A useful ML screen must be validated against an independent diagnostic reference, such as ADOS-2 or multidisciplinary clinical consensus, in a population with realistic prevalence. The label prevalence here (68.5%) is far above the roughly 3% of eight-year-olds identified with ASD [2], and far above any plausible prevalence in unselected toddler screening. As Table 5 shows, PPV at low prevalence is sensitive to small changes in specificity. False positives and false negatives both carry costs. Missed cases delay intervention, while false positives add family burden and demand on assessment services. Operating points should therefore be chosen for the intended setting. Future work should use clinician-confirmed outcomes, report calibration, and test performance across age, sex and ethnic groups. Instrument fusion [8] and item reduction [6, 7] are more promising uses of ML than re-learning a published threshold.

Limitations. First and most important, all numerical results come from a synthetic benchmark (Section 3.2), not from the public dataset. The generator reproduces the labelling rule and class balance but assumes items that are independent given latent status, and demographic effect sizes chosen by us. Results for demographic features and non-linear models in particular may differ on the public data. The qualitative conclusion about the label follows from its definition and does not depend on this choice. Second, the public dataset itself is self-selected through a mobile application, has no clinical confirmation of diagnosis, and its descriptor articles are subject to an expression of concern [13]. Third, hyper-parameters were not tuned. This avoids optimistic bias, but it may disadvantage some models on harder labels. Fourth, the E2 splits overlap, and the E1 test set is small (66 negatives), which limits the precision of specificity estimates.

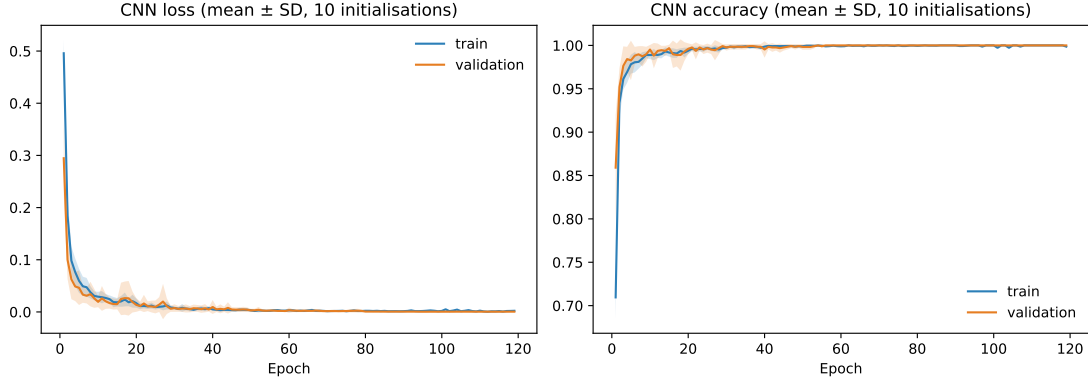


Figure 7: CNN learning curves on the E1 split: mean $\pm$ SD over 10 initialisations. Runs end at different epochs, so later epochs average fewer runs.

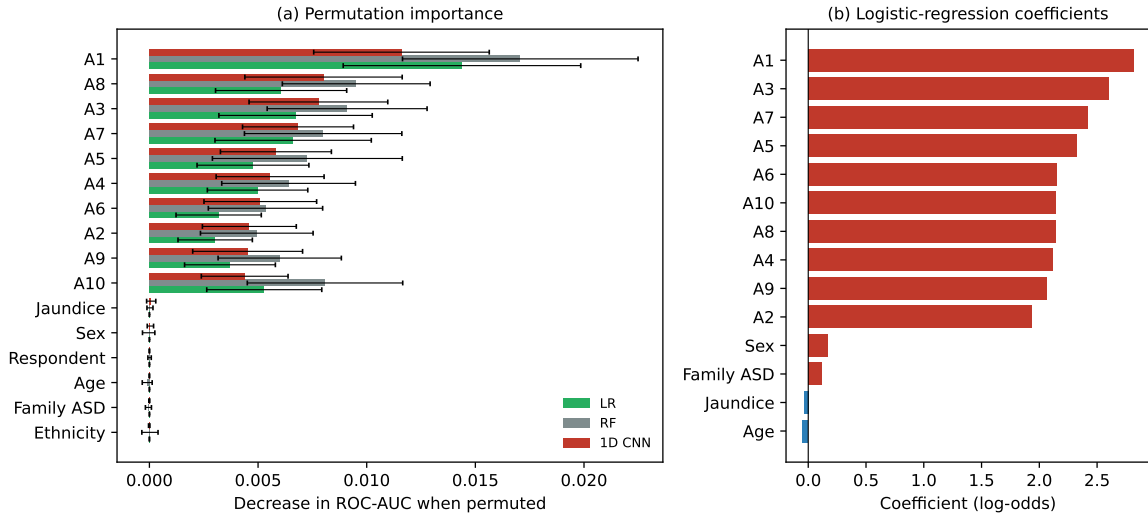


Figure 8: (a) Permutation importance (mean decrease in test ROC-AUC $\pm$ SD over 5 splits $\times$ 30 permutations). (b) Logistic-regression coefficients on the E1 split (items, binary history variables and scaled age).

6. Conclusion

We ran a leakage-controlled comparison on a synthetic benchmark constructed to reproduce the Q-CHAT-10 toddler dataset’s schema, class balance and label rule. Applying the published rule directly reproduces the threshold-defined screening label by construction. Logistic regression matches it, and a 1D CNN comes close but does not improve on it, with little sensitivity to the order of its input columns. For this label, the ten items contain essentially all the information needed to reproduce it. This says nothing about the value of other attributes for clinical endpoints. Even an error-free model leaves wide uncertainty about predictive value at population prevalence. Near-perfect accuracy on this benchmark therefore shows that the scoring rule is learnable, not that a model is clinically valid. Re-running the released pipeline on the public data, and above all evaluating models against clinician-confirmed diagnoses, are the necessary next steps.

CRedit authorship contribution statement

Subrahmanyam Tanala: Conceptualization, Methodology, Software, Formal analysis, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude Code (Anthropic) to draft software, run experiments and draft the manuscript text. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article. The released code and its outputs, not the AI-assisted workflow, are the authoritative record of the computational results. Every reported number can be regenerated independently with the command in Appendix A.

Data and code availability

The public toddler screening dataset is described in [11]. No data from it are included in, or were used for, the results of this paper. The synthetic benchmark, its generator, the model code, the scripts that produce every table and figure, a machine-readable results file (paper/generated/results_summary.json) and pinned package versions (requirements-lock.txt) are available at <https://github.com/subrahmanyamtanala-rgb/-autism-spectrum-disorders-classification-model>.

Ethics statement

Because the experiments used only author-generated synthetic data, no human-participant research was conducted in this study. The public dataset whose structure the benchmark reproduces is distributed as anonymised records without direct identifiers. Its descriptor articles [11] do not describe the consent or ethical-approval procedures of the original data collection, and those articles are subject to an expression of concern [13].

Appendix A. Reproducibility: seeds and commands

Table A.1: Random seeds used in each experiment.

Component	Seed(s)
Synthetic benchmark generator	42
E1 train/test split and models	42
E2/E3 holdout splits (and model seeds)	0–19
E4 permutation generator	0
E5 CNN re-initialisations	1000–1009
E6 splits / column permutations	0–4 / 100–102

Commands used to produce this paper:

```
pip install -r requirements-lock.txt
python experiments.py --synthetic
cd paper
pdflatex main.tex; pdflatex main.tex
```

Running with `--data <csv>` in place of `--synthetic` applies the same pipeline to the public dataset.

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